In [2]:

```
pip install pandas numpy matplotlib seaborn scikit-learn
```

```
Requirement already satisfied: pandas in c:\users\anura\anaconda3\lib\site-packages (2.
2.2)
Requirement already satisfied: numpy in c:\users\anura\anaconda3\lib\site-packages (1.2
6.4)
Requirement already satisfied: matplotlib in c:\users\anura\anaconda3\lib\site-packages
(3.9.2)
Requirement already satisfied: seaborn in c:\users\anura\anaconda3\lib\site-packages (0.
13.2)
Requirement already satisfied: scikit-learn in c:\users\anura\anaconda3\lib\site-package
s (1.5.1)
Requirement already satisfied: python-dateutil>=2.8.2 in c:\users\anura\anaconda3\lib\si
te-packages (from pandas) (2.9.0.post0)
Requirement already satisfied: pytz>=2020.1 in c:\users\anura\anaconda3\lib\site-package
s (from pandas) (2024.1)
Requirement already satisfied: tzdata>=2022.7 in c:\users\anura\anaconda3\lib\site-packa
ges (from pandas) (2023.3)
Requirement already satisfied: contourpy>=1.0.1 in c:\users\anura\anaconda3\lib\site-pac
kages (from matplotlib) (1.2.0)
Requirement already satisfied: cycler>=0.10 in c:\users\anura\anaconda3\lib\site-package
s (from matplotlib) (0.11.0)
Requirement already satisfied: fonttools>=4.22.0 in c:\users\anura\anaconda3\lib\site-pa
ckages (from matplotlib) (4.51.0)
Requirement already satisfied: kiwisolver>=1.3.1 in c:\users\anura\anaconda3\lib\site-pa
ckages (from matplotlib) (1.4.4)
Requirement already satisfied: packaging>=20.0 in c:\users\anura\anaconda3\lib\site-pack
ages (from matplotlib) (24.1)
Requirement already satisfied: pillow>=8 in c:\users\anura\anaconda3\lib\site-packages
(from matplotlib) (10.4.0)
Requirement already satisfied: pyparsing>=2.3.1 in c:\users\anura\anaconda3\lib\site-pac
kages (from matplotlib) (3.1.2)
Requirement already satisfied: scipy>=1.6.0 in c:\users\anura\anaconda3\lib\site-package
s (from scikit-learn) (1.13.1)
Requirement already satisfied: joblib>=1.2.0 in c:\users\anura\anaconda3\lib\site-packag
es (from scikit-learn) (1.4.2)
Requirement already satisfied: threadpoolctl>=3.1.0 in c:\users\anura\anaconda3\lib\site
-packages (from scikit-learn) (3.5.0)
Requirement already satisfied: six>=1.5 in c:\users\anura\anaconda3\lib\site-packages (f
rom python-dateutil>=2.8.2->pandas) (1.16.0)
Note: you may need to restart the kernel to use updated packages.
```

In [14]:

```python
import pandas as pd

# Load the dataset
df = pd.read_csv(r"C:\Users\anura\OneDrive\Desktop\internship\tested.csv")
df.head()
```

Out[14]:

	PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Em
0	892	0	3	Kelly, Mr. James	male	34.5	0	0	330911	7.8292	NaN	
1	893	1	3	Wilkes, Mrs. James	female	47.0	1	0	363272	7.0000	NaN	

	PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Em
				(Ellen Needs)								
2	894	0	2	Myles, Mr. Thomas Francis	male	62.0	0	0	240276	9.6875	NaN	
3	895	0	3	Wirz, Mr. Albert	male	27.0	0	0	315154	8.6625	NaN	
4	896	1	3	Hirvonen, Mrs. Alexander (Helga E Lindqvist)	female	22.0	1	1	3101298	12.2875	NaN	

In [16]:

```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 418 entries, 0 to 417
Data columns (total 12 columns):
 #   Column       Non-Null Count  Dtype  
---  ------       --------------  -----  
 0   PassengerId  418 non-null    int64  
 1   Survived     418 non-null    int64  
 2   Pclass       418 non-null    int64  
 3   Name         418 non-null    object 
 4   Sex          418 non-null    object 
 5   Age          332 non-null    float64
 6   SibSp        418 non-null    int64  
 7   Parch        418 non-null    int64  
 8   Ticket       418 non-null    object 
 9   Fare         417 non-null    float64
 10  Cabin        91 non-null     object 
 11  Embarked     418 non-null    object 
dtypes: float64(2), int64(5), object(5)
memory usage: 39.3+ KB
```

In [18]:

```
df.describe()
```

Out[18]:

	PassengerId	Survived	Pclass	Age	SibSp	Parch	Fare
count	418.000000	418.000000	418.000000	332.000000	418.000000	418.000000	417.000000
mean	1100.500000	0.363636	2.265550	30.272590	0.447368	0.392344	35.627188
std	120.810458	0.481622	0.841838	14.181209	0.896760	0.981429	55.907576
min	892.000000	0.000000	1.000000	0.170000	0.000000	0.000000	0.000000
25%	996.250000	0.000000	1.000000	21.000000	0.000000	0.000000	7.895800
50%	1100.500000	0.000000	3.000000	27.000000	0.000000	0.000000	14.454200
75%	1204.750000	1.000000	3.000000	39.000000	1.000000	0.000000	31.500000
max	1309.000000	1.000000	3.000000	76.000000	8.000000	9.000000	512.329200

In [20]:

```
df.isnull().sum()
```

Out[20]:

```
PassengerId      0
Survived         0
Pclass           0
Name             0
Sex              0
Age             86
SibSp            0
Parch            0
Ticket           0
Fare             1
Cabin          327
Embarked         0
dtype: int64
```

In [22]:

```
df.head()
```

Out[22]:

	PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Em
0	892	0	3	Kelly, Mr. James	male	34.5	0	0	330911	7.8292	NaN	
1	893	1	3	Wilkes, Mrs. James (Ellen Needs)	female	47.0	1	0	363272	7.0000	NaN	
2	894	0	2	Myles, Mr. Thomas Francis	male	62.0	0	0	240276	9.6875	NaN	
3	895	0	3	Wirz, Mr. Albert	male	27.0	0	0	315154	8.6625	NaN	
4	896	1	3	Hirvonen, Mrs. Alexander (Helga E Lindqvist)	female	22.0	1	1	3101298	12.2875	NaN	

In [26]:

```
# Fill missing Age with median
df['Age'] = df['Age'].fillna(df['Age'].median())

# Fill Embarked with mode
df['Embarked'] = df['Embarked'].fillna(df['Embarked'].mode()[0])
```

In [28]:

```
# Dropping Irrelevant columns
df.drop(['Name', 'Ticket', 'Cabin', 'PassengerId'], axis=1, inplace=True)
```

In [30]:

```
df['Sex'] = df['Sex'].map({'male': 0, 'female': 1})
df['Embarked'] = df['Embarked'].map({'S': 0, 'C': 1, 'Q': 2})
```

In [32]:

```
from sklearn.model_selection import train_test_split

X = df.drop('Survived', axis=1)
y = df['Survived']

X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=42
```

In [34]:

```
from sklearn.ensemble import RandomForestClassifier

model = RandomForestClassifier(n_estimators=100, random_state=42)
model.fit(X_train, y_train)
```

Out[34]:

```
▾ RandomForestClassifier
RandomForestClassifier(random_state=42)
```

In [36]:

```
from sklearn.metrics import classification_report, confusion_matrix, accuracy_score

y_pred = model.predict(X_test)

print(confusion_matrix(y_test, y_pred))
print(classification_report(y_test, y_pred))
print("Accuracy:", accuracy_score(y_test, y_pred))
```

```
[[50  0]
 [ 0 34]]
              precision    recall  f1-score   support

           0       1.00      1.00      1.00        50
           1       1.00      1.00      1.00        34

    accuracy                           1.00        84
   macro avg       1.00      1.00      1.00        84
weighted avg       1.00      1.00      1.00        84

Accuracy: 1.0
```

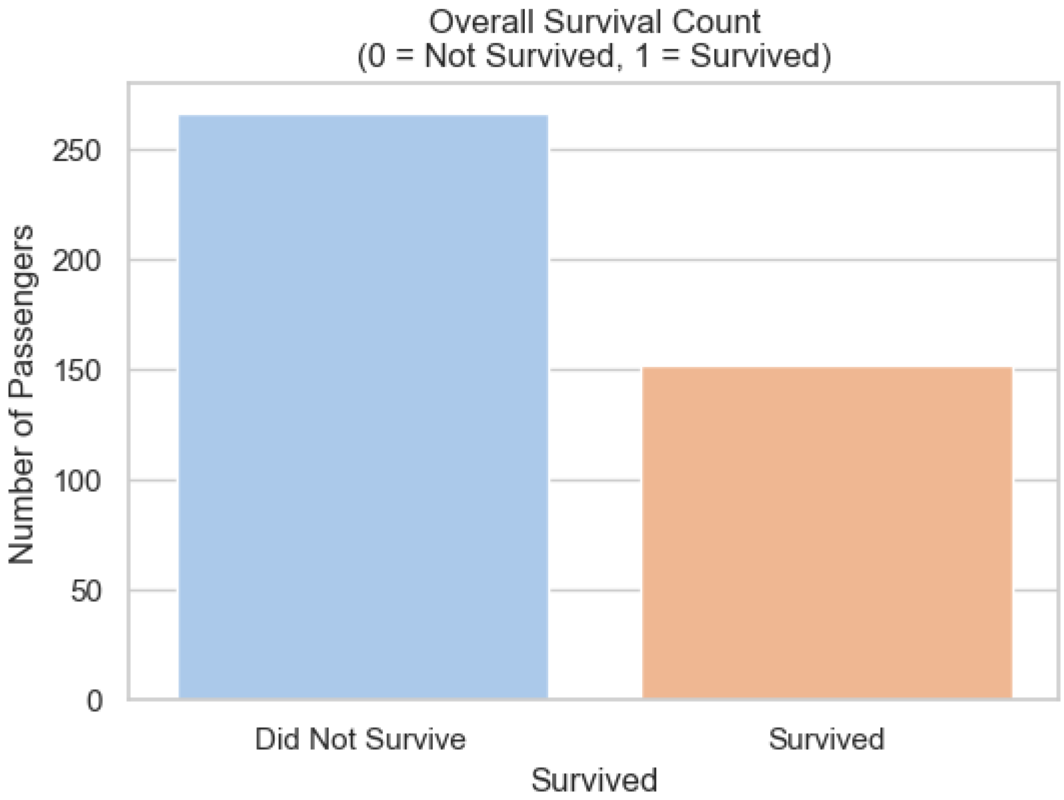
In [40]:

```
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns

# Optional: set a nicer theme
sns.set(style="whitegrid")
```

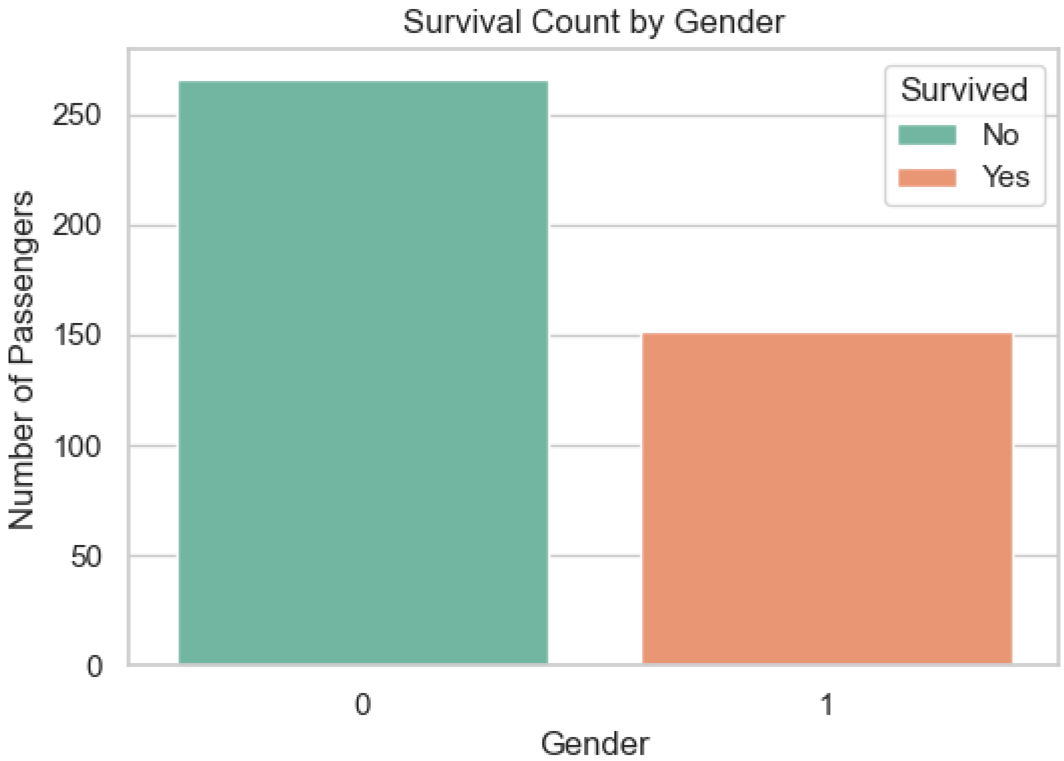
In [63]:

```
#Plot 1: Survival Count
plt.figure(figsize=(6, 4))
sns.countplot(data=df, x='Survived', hue='Survived', palette='pastel', legend=False)
```

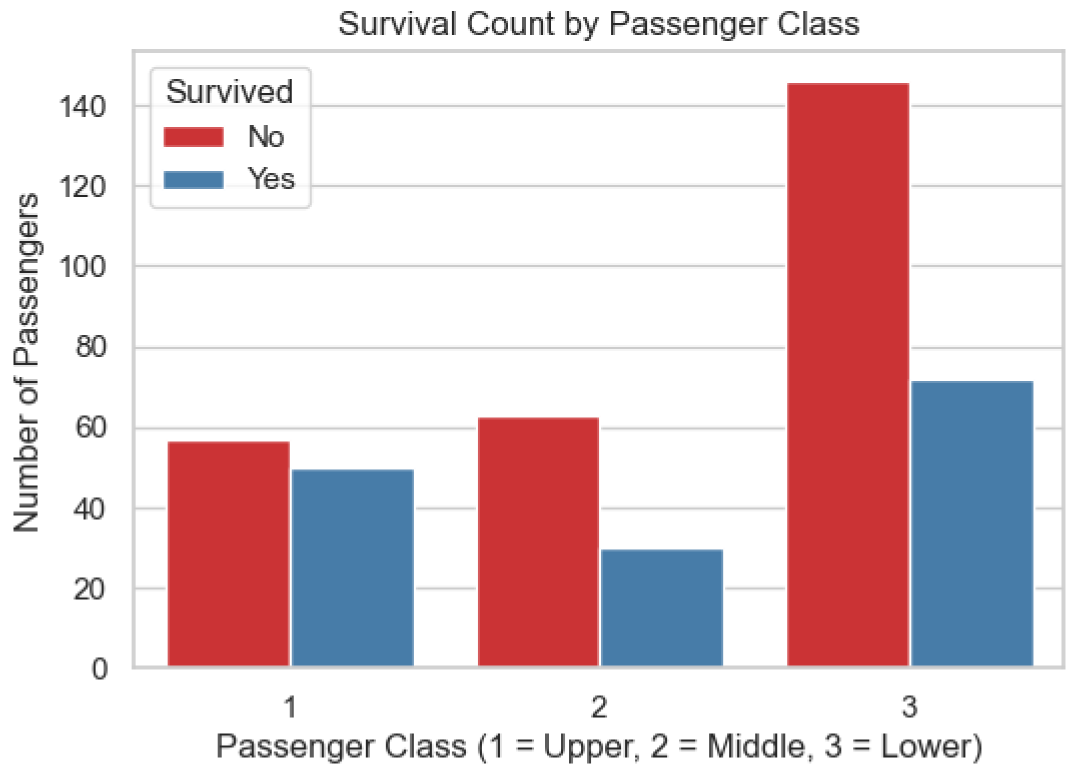
```
plt.title('Overall Survival Count\n(0 = Not Survived, 1 = Survived)')
plt.xlabel('Survived')
plt.ylabel('Number of Passengers')
plt.xticks([0, 1], ['Did Not Survive', 'Survived'])
plt.show()

# Plot 2: Survival by Gender
plt.figure(figsize=(6, 4))
sns.countplot(data=df, x='Sex', hue='Survived', palette='Set2')
plt.title('Survival Count by Gender')
plt.xlabel('Gender')
plt.ylabel('Number of Passengers')
plt.legend(title='Survived', labels=['No', 'Yes'])
plt.show()

# Plot 3: Survival by Passenger Class
plt.figure(figsize=(6, 4))
sns.countplot(data=df, x='Pclass', hue='Survived', palette='Set1')
plt.title('Survival Count by Passenger Class')
plt.xlabel('Passenger Class (1 = Upper, 2 = Middle, 3 = Lower)')
plt.ylabel('Number of Passengers')
plt.legend(title='Survived', labels=['No', 'Yes'])
plt.show()

# Plot 4: Age Distribution by Survival
plt.figure(figsize=(8, 5))
sns.histplot(data=df, x='Age', hue='Survived', multiple='stack', palette='coolwarm', bin
plt.title('Age Distribution: Survived vs Not Survived')
plt.xlabel('Age')
plt.ylabel('Count')
plt.legend(title='Survived', labels=['No', 'Yes'])
plt.show()

# Plot 5: Age vs Survival (Box Plot)
plt.figure(figsize=(6, 4))
sns.boxplot(x='Survived', y='Age', data=df, hue='Survived', palette='coolwarm', legend=F
plt.title('Age Range Distribution by Survival')
plt.xlabel('Survived')
plt.ylabel('Age')
plt.xticks([0, 1], ['Did Not Survive', 'Survived'])
plt.show()

# Plot 6: Survival by Embarkation Port
plt.figure(figsize=(6, 4))
sns.countplot(data=df, x='Embarked', hue='Survived', palette='Accent')
plt.title('Survival Count by Port of Embarkation')
plt.xlabel('Port (C = Cherbourg, Q = Queenstown, S = Southampton)')
plt.ylabel('Number of Passengers')
plt.legend(title='Survived', labels=['No', 'Yes'])
plt.show()

# Plot 7: Survival by Family Size
plt.figure(figsize=(8, 5))
sns.countplot(data=df, x='FamilySize', hue='Survived', palette='muted')
plt.title('Survival Count by Family Size')
plt.xlabel('Total Family Members Onboard (Including Self)')
plt.ylabel('Number of Passengers')
plt.legend(title='Survived', labels=['No', 'Yes'])
plt.show()
```

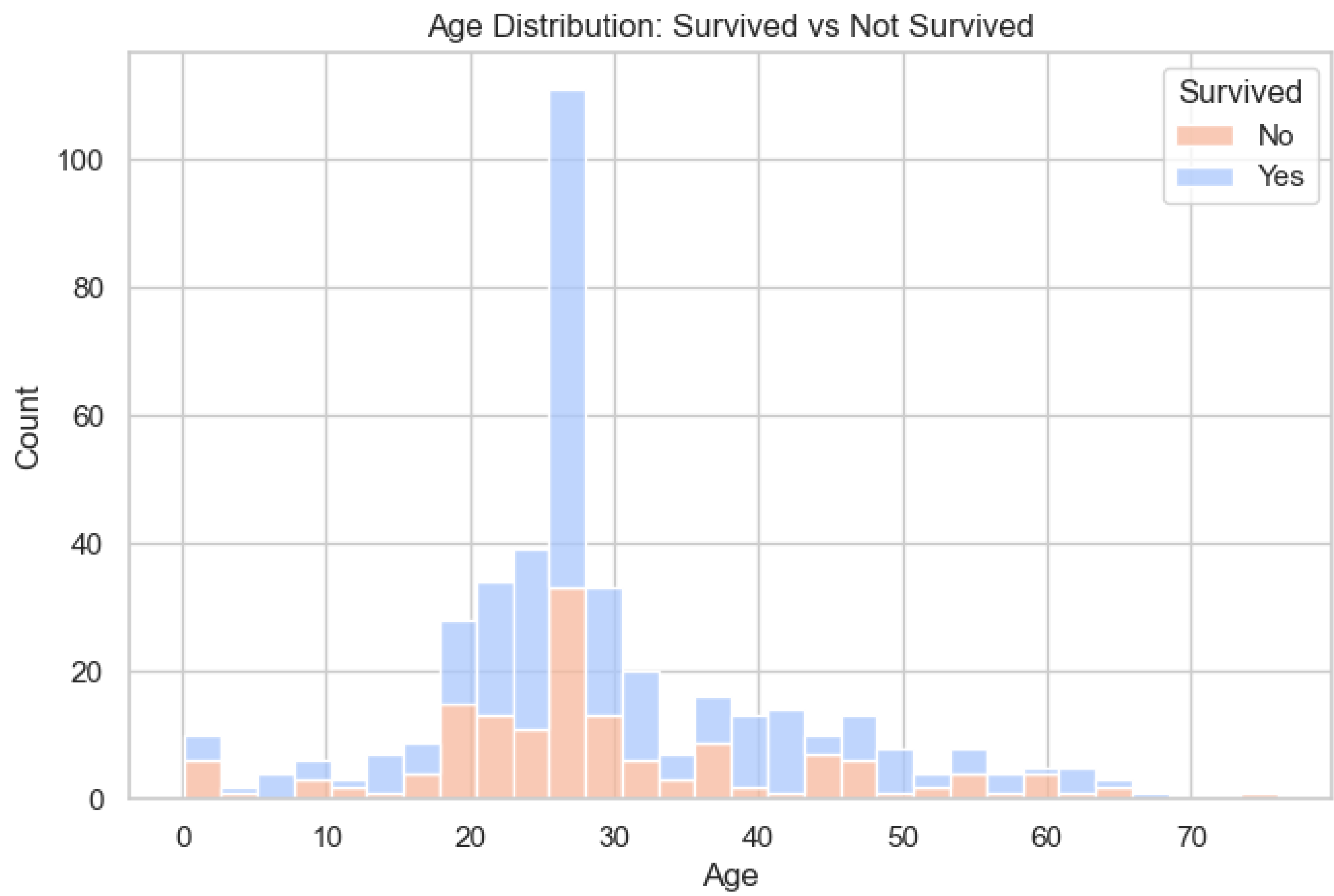


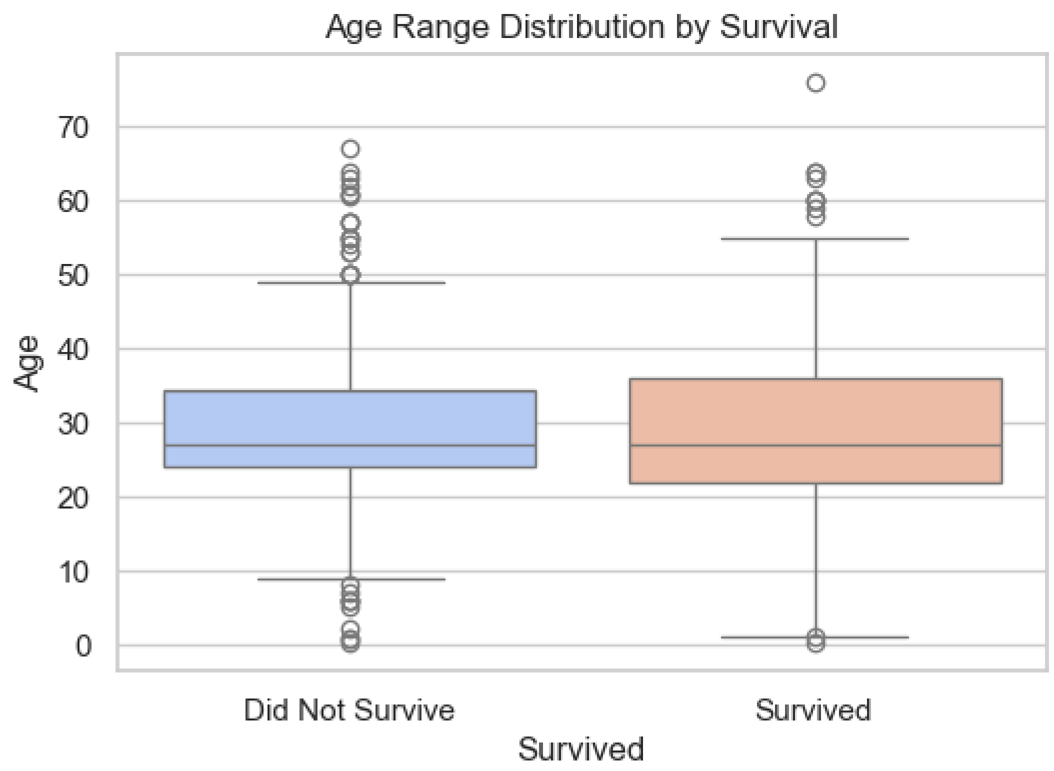
Overall Survival Count
(0 = Not Survived, 1 = Survived)
Number of Passengers
0
50
100
150
200
250
Did Not Survive
Survived
Survived

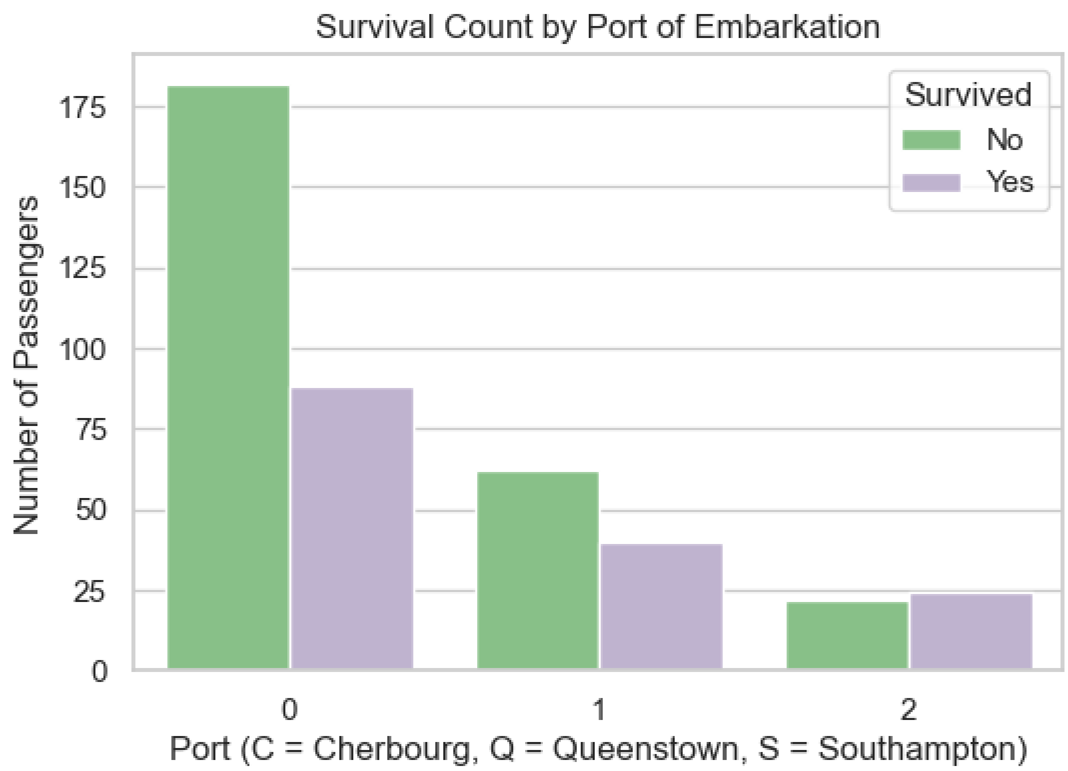


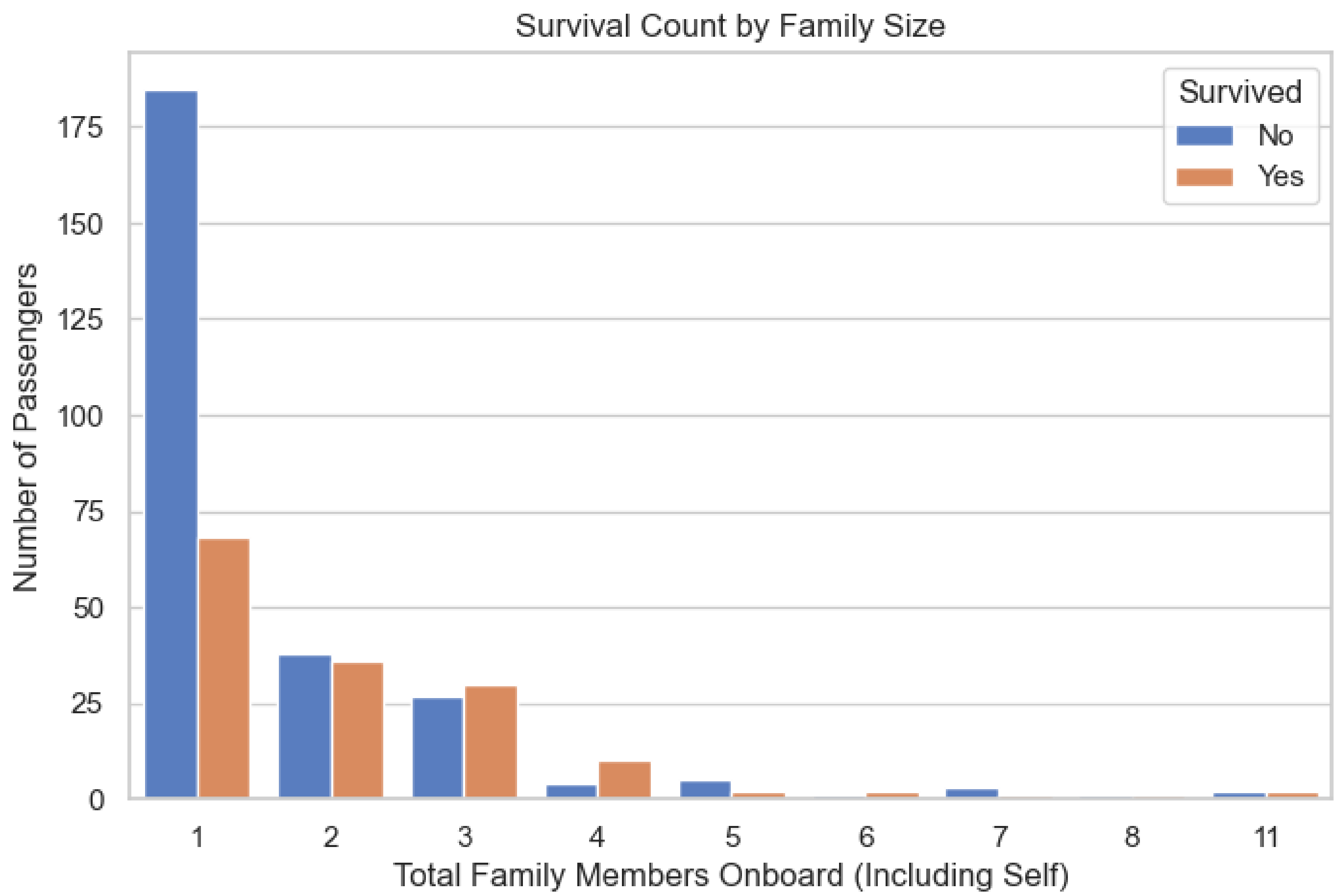
Survival Count by Gender
Number of Passengers
0
50
100
150
200
250
Survived
No
Yes
0
1
Gender


Survival Count by Passenger Class
Survived
No
Yes
Number of Passengers
0
20
40
60
80
100
120
140
1
2
3
Passenger Class (1 = Upper, 2 = Middle, 3 = Lower)


Age Distribution: Survived vs Not Survived
Survived
No
Yes
Count
0
20
40
60
80
100
0
10
20
30
40
50
60
70
Age


Age Range Distribution by Survival
Age
0
10
20
30
40
50
60
70
Did Not Survive
Survived
Survived


Survival Count by Port of Embarkation
Number of Passengers
0
25
50
75
100
125
150
175
Survived
No
Yes
0
1
2
Port (C = Cherbourg, Q = Queenstown, S = Southampton)


Survival Count by Family Size
Survived
No
Yes
175
150
125
100
75
50
25
0
Number of Passengers
1
2
3
4
5
6
7
8
11
Total Family Members Onboard (Including Self)

```
In [ ]:
```